Econ 2 - Lecture 12 - 2/23/26

Lecture Quiz # 7 Released Wednesday

↳ Due Monday 3/2

Discussion Activity #4 → Gamification!

Sign in using "passcode" on Canvas

Last class: Using $Y = AE$ to explain macroeconomy

→ Multiplier effect ⇒ Expenditure Multiplier

→ Policy Actions: Ask for help to increase AE?

Government / Policy Maker Role: Get back to $\bar{Y}$!

Fiscal Policy: $\underbrace{\text{Actions taken by Government}}_{\text{Congress / President}}$

What can government change?

1. Change G, ΔG

2. Change T, ΔT

If $Y_{08} < \bar{Y}$, what should gov't do? Increase Y

1. $\Delta G > 0$

2. $\Delta T < 0$

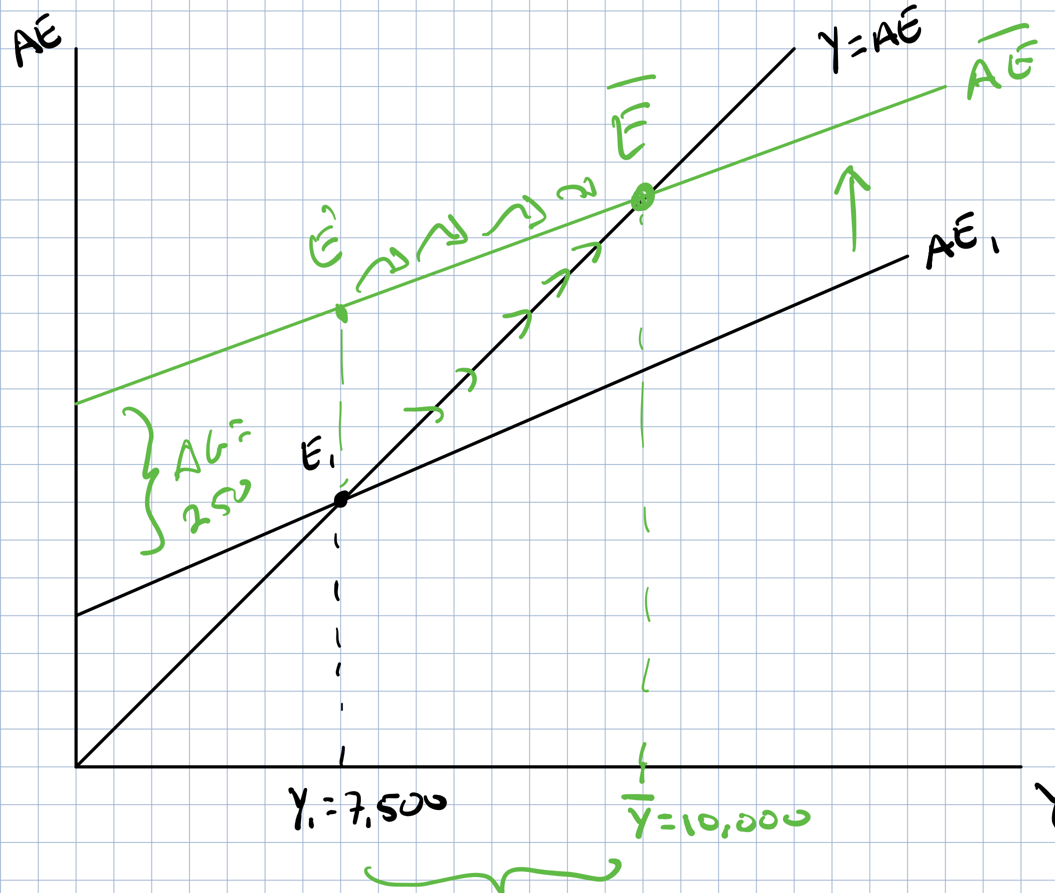
3. Do nothing → Time?

Counter-cyclical fiscal Policy → ΔG or ΔT

If $Y < \bar{Y}$ → Increase Y, If $Y > \bar{Y}$ → Decrease Y

$Y_1 = 7{,}500$, $\bar{Y} = 10{,}000$, $mpc = 0.9$

How much should we ΔG in order to get back to $\bar{Y}$?



$$\Delta Y = \frac{1}{1-mpc}\Delta G \Rightarrow 2500 = \frac{1}{1-0.9}\Delta G = 10 \times \Delta G$$

$$\Delta G = 250$$

If ΔG by 250 $\rightarrow \bar{Y} = 10{,}000$ [Easy?]

What does it take to increase G?

Non-controversial gov't project?

Bridge in Cincinnati $\rightarrow$ Hwy 71/75

$\rightarrow$ # ~~1~~ #2 infrastructure issue in US

#1: Train tracks in NYC

→ 4% of GDP crosses bridge annually

2012: Obama & R. Paul
Agree that bridge should be built

Bill proposed part of a large spending bill

McConnell → KY senator → wasteful
↳ if you want a bridge
ask!

Imagine bill is passed in timely manner
→ After bill is passed
→ Who is the builder? Bids by contractors.
One option: most expensive bid (gold plated roads)
Another option: cheapest bid (avoid this)
Last option: I got a cousin → builds bridges

Take Decades to fix big problems

Explore alternative to ΔG:

Change Taxes $\Rightarrow$ Decrease Taxes

Note: If you want $\Delta Y > 0$, $\Delta T < 0$

$\Delta Y < 0$, $\Delta T > 0$

When $mpc = 0.9$, $\Delta Y = 2500$, $\Delta G = 250$

	$\Delta G = 250$	$\Delta T = -250$
Round 1	Build Bridge for \$250 ($\uparrow G$, $\uparrow Y$ by 250)	Take-home income increases by 250 (No Bridge, no ΔY)
	$\downarrow$	$\downarrow$
Round 2	Spend $mpc \times 250$ $= 0.9 \times 250$ $= 225$ ($\uparrow Y$ by 225)	Spend $mpc \times 250$ $= 0.9 \times 250$ $= 225$ ($\uparrow Y$ by 225)
	⋮	⋮
Total	$\Delta Y = 10 \times 250$ $\underbrace{\Delta Y = 2500}$	$\Delta Y = 2500 - 250$ $\Delta Y = 2250$

Expenditure Multiplier $= 10$

Tax Multiplier $= -9$

$\Delta Y = 2250 = \text{Tax Mult.} \times \Delta T$

$2250 = \text{Tax Mult.} \times (-250)$

Tax Multiplier = fraction of expenditure Multiplier

$$= \frac{-mpc}{1-mpc}$$

If $mpc = 0.9$, Tax Multiplier $= \frac{-0.9}{1-0.9} = \frac{-0.9}{0.1}$

Tax Multiplier $= -9$

$\Delta Y = 4000$, $mpc = 0.8$, $\Delta T = ?$

$$\Delta Y = \frac{-mpc}{1-mpc} \Delta T$$

$$4000 = \frac{-0.8}{1-0.8} \Delta T = \frac{-0.8}{0.2} \Delta T = -4 \cdot \Delta T$$

$$\Delta T = \frac{4000}{-4} = -1000$$

Decrease T by 1000 → increase in Y of 4000

$\Delta T < 0 \rightarrow \Delta Y > 0$

Increase G by 1000 → increase in Y $= \frac{1}{1-0.8}(1000)$

$= 5000$

Tax Multiplier $= \left| \frac{-mpc}{1-mpc} \right| < \frac{1}{1-mpc}$ = Expenditure Multiplier

Who gets tax cut?

Imagine policy makers → only one group gets a tax cut

Option #1: Give tax cut to low-income households

Benefit: mpc is large ⇒ mpc = 0.95

→ large multiplier ⇒ $-\frac{0.95}{1-0.95} = \frac{-0.95}{0.05} = -19$

Cost / Limitation: $\Delta Y = -19 \cdot \underline{\Delta T}$

Income = 30K → Taxes = \$4K/yr

Max ΔT = \$4K

Change in Y could be limited

Option #2: Give tax cut to rich/high-income

Benefit: Large tax base

\$1 million/yr salary → Pay \$350K/yr in federal Taxes

ΔT can be very large → small percentage change in T ⇒ ΔT is large

Small Business Tax Cuts → in return for hiring → Reduce Unemployment

Cost: mpc could be ϕ

$\Delta Y = \phi\,(\Delta T < 0) = \phi$

Recap: Found where $Y = AE$

If $Y \neq AE \rightarrow$ counter-cyclical fiscal policy $= \Delta G, \Delta T$

Get back to $\bar{Y} =$ full-employment Y

Need Congress/President to agree

Biggest constraint of fiscal Policy → Costly!

When there is a big shock to $Y \rightarrow \Delta G, \Delta T$ happen

Great Financial Crisis

Bailout Bill: Loan by US Treasury → \$700 billion

→ US Treasury made \$15 billion

Stimulus Package:

American Reinvestment & Recovery Act (ARRA)

$\Delta G = 500$ billion, $\Delta T = -300$ billion

$\Delta Y = ?$ mpc during GFC?